

# The Art of Queueing

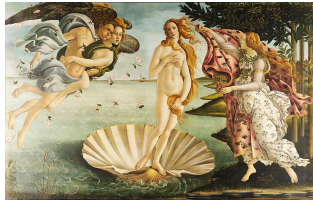
Reinforcement learning for visitor flow at the Uffizi

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# The rooms everyone queues for

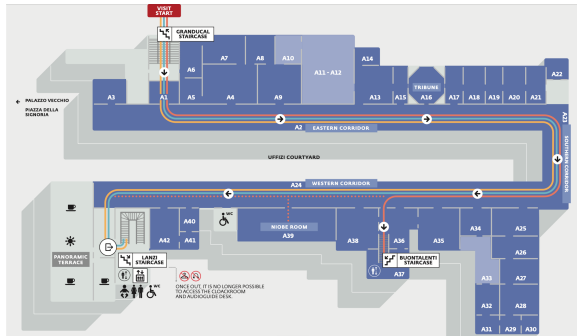


Botticelli's *Primavera* and *Birth of Venus* (room A11/A12); Leonardo's *Annunciation* and *Adoration of the Magi* (A35); Raphael's *Madonna del Cardellino* with Michelangelo's *Doni Tondo* (A38). Almost every visitor comes for these few rooms.

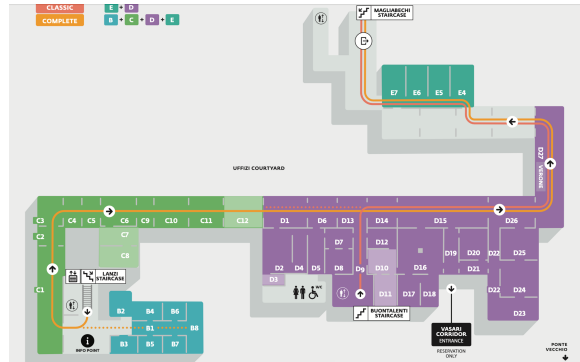
# Too many people for a few rooms

The Uffizi takes ~5,000 visitors a day and almost all want those few rooms; the other ~90 sit near-empty. The crowd is 60% Instagram, 30% normal, 10% art lovers. Not too many people for the building, too many for a few rooms at the same moments.

## Second floor (the three gated masterpieces)



## First floor (Caravaggio, free-flow)



# Two streams, one project

## Stream 1: crowd simulator.

The whole 60/30/10 population, a full day.

- Museum *as-is* vs a Pigovian redesign.
- *Director gate*: welfare up, revenue not down.
- Result: welfare +31%, revenue preserved.

## Stream 2: RL on top (our focus).

Take the redesigned museum as given.

- What does the *optimal* visitor *do*?
- When to book, which masterpieces, how to pace.
- Under crowd uncertainty.

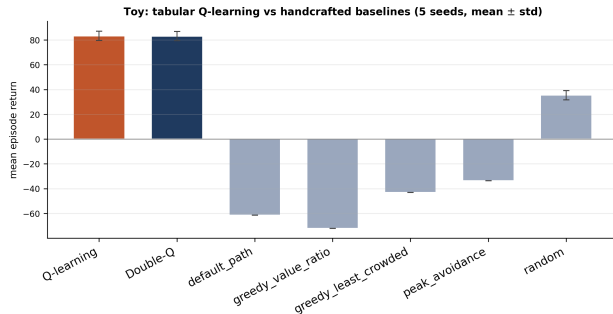
*The gate passes first; only then do we study the optimal visit.*

# Foundations: a tabular proof of concept

A 12-room toy with deterministic crowd waves, small enough for an exact table. *Q-learning* and *Double-Q*.

It learns *temporal arbitrage*: enter a masterpiece in the crowd valley, not at the peak. It is never told the schedule.

*Honest result (5 seeds)*: Q and Double-Q tie (83.3 vs 83.1). On a deterministic toy there is no overestimation for Double-Q to fix.



# Scaling up: deep RL and action masking

The real museum breaks the table: 98 rooms,  $\sim 100$  actions, a  $\sim 400$ -dim state. So we switch to neural networks and *MaskablePPO*, with unmasked PPO and DQN as controls.

## Action masking: hand the agent the rulebook

The museum is a graph, so most actions are illegal at each step. We force every illegal action to zero probability before the softmax. No gradient is wasted learning “which moves exist”; it all goes into the real decision.

*It will turn out to be decisive exactly where the decision is hardest.*

# The decisive reframe: a booking problem

We spent months on a *navigation* agent. It was brittle and the worry grew: is routing with a map even a real RL problem? *It is not* (a visitor has a map and a guide).

The genuine, uncertain decision is the *booking*:

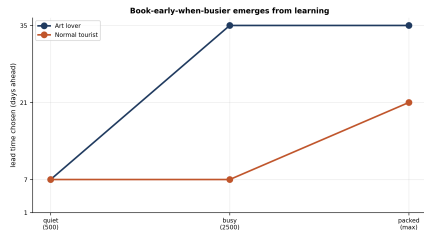
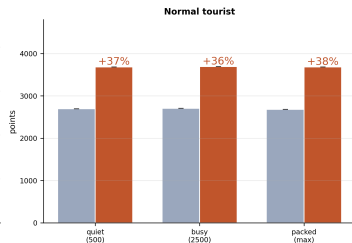
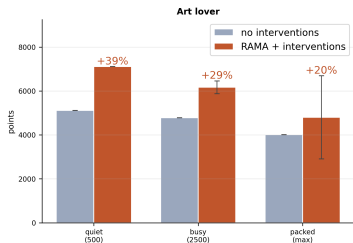
- *baseline (as-is museum)*: no booking, just walk the crowds, the museum as it runs today;
- *intervened (RAMA)*: masterpieces gated by booked time-slots, so the decision becomes *lead time*  $\in \{1, 7, 21, 35\}$  days, then pace to each window.

A crowd-dependent *fill curve* makes good slots sell out on busy days, so booking late at a packed museum misses masterpieces.

*Lesson: choosing the MDP is more than half the work.*

# Result: book early when busier and it pays

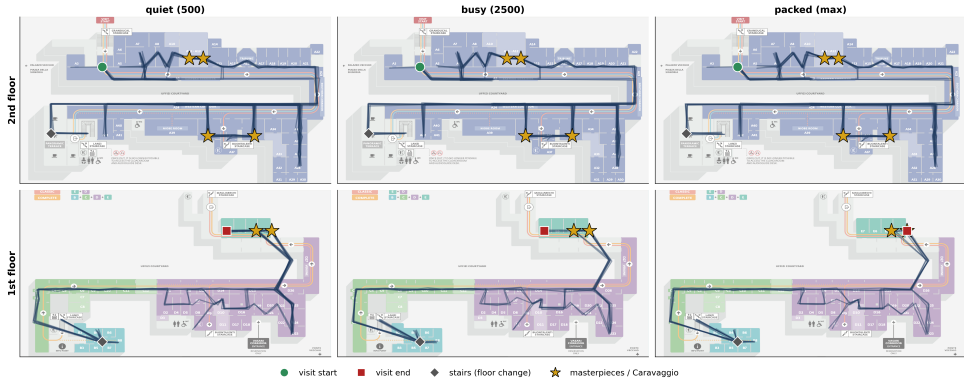
RAMA booking: intervened vs matched baseline (5 seeds, mean  $\pm$  std; 3/3 masterpieces secured)



Both profiles beat the matched no-booking baseline (5 seeds, mean  $\pm$  std): **+20% to +39%** (art), **+36% to +38%** (tourist). Lead rises with the crowd: book-early emerges from learning.

# What a visit looks like, on the real floor plan

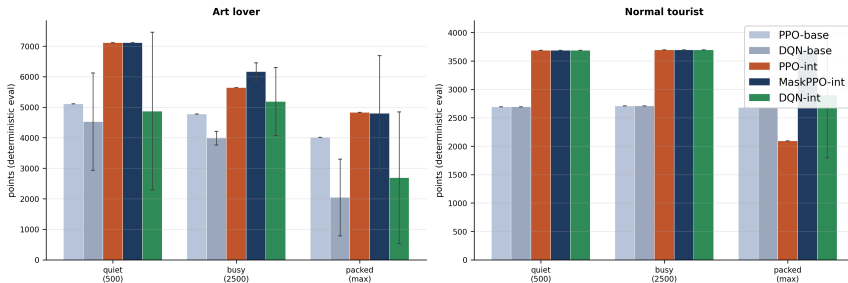
Art lover walking routes (RAMA + interventions) -- one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



*Art lover, eight rollouts per panel, across the three crowds. One continuous visit: enter at A1, sweep the second-floor masterpieces (gold stars), drop to Caravaggio on the first floor, exit. Every move is a one-hop neighbour; all three masterpieces are hit inside their booked windows.*

# Which algorithm wins and why masking matters

Algorithm comparison: value-based (DQN) vs policy-gradient (PPO / MaskablePPO), baseline vs intervened (5 seeds, mean  $\pm$  std)



- *MaskablePPO* is best or tied everywhere, stable across seeds.
- *Masking is decisive at the hardest cell*: at the packed tourist, unmasked PPO collapses to 2093 (below the 2680 baseline) while MaskablePPO holds 3685.
- *DQN* is crowd-fragile and high-variance (baseline 4527  $\rightarrow$  2044): its max over noisy crowd values overestimates, the bias the toy never had.

# The journey was the project: 1, an unlearnable reward

**S** *Symptom.* The agent never left the museum: it squatted in a room until the day ended. Zeroing all reward for failing to exit made it *worse*.

**D** *Diagnosis.* An all-or-nothing terminal wipe is *flat* across almost every policy, so the gradient is zero. Nothing points toward the door.

**F** *Fix.* A *dense* egress signal: per-step pressure that grows as closing nears, a finite (not infinite) no-exit penalty, an exit bonus.

**L** *Lesson.* Sparse all-or-nothing rewards are often unlearnable. Shape a slope the optimiser can climb.

## 2: the agent could not see what it had to decide on

**S** *Symptom*. “Stay until satisfied, then move on” never stabilised; the agent over- and under-dwelt rooms at random.

**D** *Diagnosis*. The right action depended on *how much value it had already extracted here*, which was *not in the observation*, a hidden variable (a POMDP).

**F** *Fix*. Add a per-room *appreciation-progress* value to the state, so “have I seen enough?” is something the policy can read.

**L** *Lesson*. The state must contain everything the optimal action depends on.

### 3: reward shaping that backfired

**S** *Symptom.* A “boredom” penalty produced *looping*; a harsh crowd penalty produced a *fake art lover* that skipped crowded masterpieces.

**D** *Diagnosis.* Every term you add is a new objective the optimiser games (reward hacking): circling beat exiting; a crowded Botticelli scored worse than skipping it.

**F** *Fix.* *Minimal* shaping: a gentle crowd discount (a crowded masterpiece still beats skipping), drop the boredom penalty, let satiation carry “move on”.

**L** *Lesson.* Stress-test every reward term for the perverse incentive it creates.

## 4: the do-nothing trap and the trick that cracked book-early

**S** *Symptom.* Given a free “decline” button, *both* profiles declined all masterpieces. Even without it, book-early was only *half*-learned.

**D** *Diagnosis.* Declining is immediately safe while booking pays later (a do-nothing trap); and the booking payoff is hundreds of steps away, so its gradient is faint (credit assignment).

**F** *Fix.* Treat “always books” as a *type trait* and add an *immediate off-pace penalty*, a local proxy correlated with the distant cost.

**L** *Lesson.* A small immediate signal that tracks a distant reward is what makes a long-horizon lesson learnable. The single most decisive trick.

# What five seeds revealed

A single run can be lucky, so we retrained every cell under *five* seeds and report the spread.

- *Tourist*: rock-stable at every crowd.
- *Art lover*: the gain degrades as the museum fills ( $+39 \rightarrow +29 \rightarrow +20\%$ ).
- *The honest catch*: at *max crowd* the art lover is *bimodal*, four of five seeds reach  $\sim +41\%$  but one *collapses*, so it averages  $+20\%$  with a wide band. We report the spread, not the lucky seed.

*Reporting variance honestly is part of the result.*

# What worked best and worst

## Worked best

- *Action masking*: the single biggest lever.
- *Reframing to booking*: the right MDP.
- *An immediate proxy* for the delayed reward.
- *Exposing appreciation progress*: Markov again.
- *Dense egress shaping*: a learnable exit.

## Worked worst

- *All-or-nothing exit wipe*: zero gradient.
- *Free decline button*: a do-nothing trap.
- *Harsh crowd penalty*: a fake art lover.
- *Boredom penalty*: a looping incentive.
- *Greedy eval* of a stochastic policy.

# Conclusion and what is debatable

- *Result:* the optimal art lover and tourist *book early, earlier when busier*, secure all three masterpieces, beat the no-booking baseline by +20% to +39% (art) and +36% to +38% (tourist).
- *The journey was the project:* a chain of instructive failures, each a named RL lesson, unlearnable reward, partial observability, reward hacking, do-nothing trap, credit assignment.
- *Honestly debatable:* the 60/30/10 split, the dwell and crowd-density calibration and room positioning are modelling choices (the floor-plan maps let you judge them); the fill curve is calibrated, not booking data; the as-is museum abstracts away the real entry choice (pre-book and skip the queue, or pay less and queue).
- *Next:* the Vatican Museums, a larger topology and real booking traces.

# Thank you.

Questions?

*Code, figures and a plain-language guide to every algorithm:*  
`github.com/mocl20/Uffizi`